

RADIO IMPLEMENTED IN SOFTWARE

Unlike hardware-based radio systems, software-defined radio can be reconfigured to suit a wide range of applications. This is one big advantage of this technology evoking interest from the telecom industry



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Traditionally, radio communication systems consist of hardware blocks only. These hardware blocks perform specified functions which cannot change during operation. Some radio systems do use programmable digital signal processors (DSPs), but their possible functions are only speech encoding and coordination of chip rate, symbol rate and bit rate coprocessors. The DSP in a system does not typically participate in computing-intensive tasks.

Addressing the radio systems composed of hardware blocks, engineers must focus carefully on the function design, i.e., logic block design. Hard-

ware-based systems ordinarily operate provided the design and the manufacture are appropriate.

Advanced mobile systems, especially high-performance 3G systems, are expected to have over two million logic gates to implement physical-layer processing. For such a system, each hardware block requires complex chip designs. If we consider multiple communication systems or multiple functions for a single system, the computationally intensive signal processing algorithms and high data rates associated with these systems necessitate dedicated hardware implementation of some portions of the signal processing chain. But, allocating separate hardware resources for each of the functions

would increase the silicon area, complicate design validation and compatibility, and also increase the cost.

Software-defined radio (SDR) takes a completely new approach to communications system design. Let's see how.

What's an SDR?

Joe Mitola coined the term 'software radio' in 1991 to refer to the class of reprogrammable or reconfigurable radio.

Communication systems based on software platform can be dynamically reconfigured. This allows efficient reuse of the silicon area and dramatically reduces time to market through software modifications instead of hardware redesigns.

So with the SDR technology, it is possible to produce a communication system that changes operations depending on the software loaded into it. As a transmitter, it analyses and characterises the available transmission channel, probes the propagation path, constructs an appropriate channel modulation, guides the transmit beam, selects the appropriate power level and then transmits. As a receiver, it recognises the mode of incoming transmission, adaptively nullifies interference, estimates the dynamic properties of the desired signal multi-path, coherently combines them, equalises, decodes and corrects errors to receive the signal with the lowest bit-error rate (BER).

Fig. 1 shows the block diagram of an ideal software radio transceiver. The digital-to-analogue and analogue-to-digital converters at transmit/receive